Country	Student No.	Experiment No.	Page No.	Total Pages

ANSWER FORM

PART A

1. State the method of your choice and sketch the experimental set up of the method: **[*0.5 pts*]**

2. Write down the expression relating the measurable quantities in your chosen method: **[*0.5 pts*]. State all the approximations used in obtaining this expression [1.0 pts].**

3. Collect and organize the data in the following orders : physical quantities, values, units **[*1.0 pts*]**

4. Indicate the quality of the calibration by showing the plot relating two independently measured quantities and mark the range of validity. **[*0.5 pts*]**

5. Determine the smallest unit of the arbitrary scale in term of mm and its estimated error induced in the measurements. **[*1.5 pts*]**

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PART B

1. Sketch of the experimental set up. **[*1.0 pts*]**

2. Derive the following expression:

$$I\,\Delta t = \frac{\mathrm{e}}{\mathrm{k_B}}\frac{2P(pr^2)}{T}\Delta \mathrm{h}$$ **[*1.5 pts*]**

3. Collect and organize the data in the following format : physical quantities (value, units) **[*1.0 pts*]**

4. Determine the value of e/k_B and its estimated error **[*1.5 pts*]**